D1/D2 results: interpretation and next checks

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Bottom line

- **Red flag in FUT:** CTRL_FUT and LCC_FUT differ strongly **outside** the land-cover region, so Total_FUT cannot yet be read as a land-cover-only effect. Check this first.
- **BVOC_PD:** the fBVOC artifact is **detected and larger than the local BVOC signal**. Its signature looks like ozone from the missing compounds.
- **BVOC_FUT** is dominated by the climate-driven global rise in emissions, as designed. The land-cover part is not visible in it.
- **Other compounds:** their land-cover-driven change is negligible locally in PD, but **about as large as the ISOP+MTERP change in FUT**.

Tags: (a) read from the D1/D2 figures or case scripts · **(b)** inference · **(c)** guess. Numbers are read off the plots and are approximate.

1. Red flag: FUT members differ outside the land-cover region

About 99% of the FUT emission change from LCC − CTRL happens outside the modified cells, and the radiative far field is six times the local signal.

Evidence	PD	FUT
D2: global ΔE(ISOP+MTERP), LCC − CTRL	≈ −10 Tg yr^{-1}	**≈ −250 Tg yr^{-1}**
D2: same, modified cells only	≈ +0.45 Tg yr^{-1}	≈ −1.4 Tg yr^{-1}
Share of the change outside the modified cells	~95% (opposite sign)	**~99%**
D1 Total: far field 30°S–30°N, net contribution	≈ +0.01 W m^{-2} (not significant)	**≈ +0.25 W m^{-2}** (modified cells +0.04)
D1 Total maps: clear_SW over tropical land	mostly noise	**significant patches** (Amazon, Africa)
D1 Total maps: clear_LW	mainly boreal	**negative over almost all land, significant**

These fit together (b). LCC_FUT emits much less tropical BVOC, so less SOA and less aerosol cooling. That matches the positive far-field ari (+0.05) and cloud (+0.12 SW) terms in Total_FUT. Significant tropical clear_SW and global land clear_LW point to a **different land surface in the tropics**, not to an atmospheric response to the boreal change.

Likely cause (b/c):

- **Different spin-ups (a, case scripts):** CTRL_FUT restarts from `spinup_f19_f19-3`, LCC_FUT from `spinup_lcc_f19_f19-20260928`. CLM BGC vegetation (LAI, carbon pools) is prognostic, so the two can be in different states anywhere.
- **Other surfdata fields:** `modified_cells` only compares `PCT_NAT_PFT` (a). Differences in PCT_CROP, PCT_NATVEG or other fields between `simyr2000` and the LPJ-GUESS file would not show there (c).

Not affected: the BVOC contrasts compare members from the same spin-up (LCC and LCC_fBVOC both use `spinup_lcc…-20260928`) (a).

Possibly also in PD (b): the D2 global-versus-local mismatch suggests a smaller version of the same issue, although the PD radiative far field is clean.

2. BVOC_PD: the artifact is detected and matters

The far-field signal is about 6–9× the local BVOC signal, so the global BVOC_PD number is dominated by the artifact ("detected, MATTERS" in D1 terms).

Term (net), W m^{-2} of global mean	Modified cells	30°S–30°N	90°S–30°S
total	≈ +0.007	**≈ +0.045** (interval excludes 0)	**≈ +0.022**
clean clear-sky (mostly LW)	≈ 0	**≈ +0.026**	**≈ +0.015**
cloud	≈ +0.005	+0.02, not significant	≈ 0

- **Signature consistent with ozone (b).** LCC has the other compounds and LCC_fBVOC does not, so LCC should have more tropospheric O_3. This fits the positive, coherent, stippled clear-sky LW over the tropics and southern oceans, and the small positive clear-sky SW over oceans. Tropospheric CH_4 or water vapour may add a little (c).
- **The local BVOC_PD signal is very small.** That agrees with D2: land cover changes ISOP+MTERP in the modified cells by only ≈ +0.45 Tg yr^{-1}.

3. BVOC_FUT: dominated by climate change

BVOC_FUT as defined measures mainly the BVOC effect of climate change: mostly tropical SOA, plus O_3 (b).

- **D2, LCC_FUT − CTRL_PD (global, mass):** ISOP+MTERP ≈ +485 Tg yr^{-1}, others ≈ +565 Tg yr^{-1}.
- **D1 far field, 30°S–30°N:** net ≈ −0.44 W m^{-2} (cloud ≈ −0.36, ari ≈ −0.14), against ≈ −0.05 from the modified cells.
- **O_3 signature:** positive clear-sky LW everywhere and positive clear-sky SW over Antarctic sea ice (b).
- **Separating the land-cover part** needs CTRL_FUT_fBVOC, and only after the CTRL_FUT state issue (section 1) is resolved.

4. Total_PD: clean far field, coherent local signal

The modified cells contribute about **+0.03 W m^{-2}** (global-mean equivalent): forest albedo warming, partly offset by clouds (b).

Term (modified cells)	Contribution, W m^{-2} of global mean
Clean clear-sky SW (albedo)	≈ +0.06
Cloud SW	≈ −0.025
Clean clear-sky LW (warmer surface)	≈ −0.013
Net	**≈ +0.03**

The far field is not significant but noisy (tropical cloud SW interval ≈ ±0.1); nudging does not constrain clouds much. Report the local contribution and the far field separately, because the global-mean Total_PD is dominated by far-field noise.

5. D2: how big are the other compounds?

The others are about a quarter of all MEGAN carbon globally in PD, and their land-cover-driven change matters in FUT but not in PD.

Q1 – share of all MEGAN emissions in CTRL:

Weighting	PD global	PD modified cells	FUT global	FUT modified cells
Carbon	23.5%	38%	32%	44%

Weighting	PD global	PD modified cells	FUT global	FUT modified cells
OH reactivity	6%	14.5%	11%	21%

In carbon terms the share is substantial, which explains why removing them everywhere (the D1 artifact) is visible. Global totals look plausible for MEGAN2.1 (b).

Q2 – land-cover-driven change in the modified cells (mass Tg yr^{-1} / carbon Tg C yr^{-1} / OH reactivity Tmol CO-eq yr^{-1}):

Period	ISOP+MTERP	Others
PD	+0.45 / +0.39 / +2.2	≈ 0 (not significant, interval ≈ ±0.4)
FUT	−1.4 / −1.2 / −3.2	**−4.4 / −2.3 / −2.8**

- **PD:** an ISOP/MTERP-only BVOC definition misses little locally.
- **FUT:** the others' change is comparable to or larger than ISOP+MTERP in carbon and OH reactivity, mostly oxygenated VOC, then CO and other hydrocarbons. Section 1 may contaminate this too, so recheck after the state check.
- Read `others_share_of_gross_change` in the table: the figure can hide groups that cancel each other.

6. What this means for the fBVOC decisions

A fix that includes all compounds (O1 or O2-lite) is needed, but the CTRL vegetation state must be resolved first (b).

Question	Answer from D1 + D2 (b)
Does the fBVOC artifact matter?	**Yes** for global BVOC numbers (PD, through O_3); less so for the local signal
Are the other compounds negligible?	In PD locally, yes; **in FUT, no**
Then	Use **O1 or O2-lite**, but **first resolve section 1**: O1's BGP term (LCC_fBVOC − CTRL_fBVOC) would inherit any CTRL vegetation-state difference

7. Next checks

1. **Land state, LCC − CTRL, PD and FUT:** maps and tropical/global means of `TLAI`, `TOTVEGC`, `GPP`, year by year, to see both the difference and any drift. A large tropical ΔTLAI confirms section 1.
2. **Surfdata:** compare every variable between `simyr2000` and the LPJ-GUESS file, not only `PCT_NAT_PFT`.
3. **Spin-up lineage:** how long each spin-up ran and from which land state it started (CTRL_PD, CTRL_FUT, LCC). The `check-simulation-equilibrium` diagnostics may already answer this.
4. **D2 table:** read the `tropics` rows of `D2_q2.csv` to locate the FUT emission difference.

If check 1 shows the vegetation differs, CTRL_FUT (and maybe CTRL_PD) must be rerun from a spin-up consistent with the LCC family. Total_FUT and the BGP term should wait until then.